

Day 03: Regression with `scikit-learn`

Predicting continuous values

MSU REU Machine Learning Short Course

Learning Goals for Today

By the end of this activity, you'll be able to:

1. **Distinguish regression from classification** — when each is the right tool
2. **Build and train a linear regression model** using the same pipeline as classification
3. **Evaluate regression models** with MSE, R^2 , and diagnostic residual plots
4. **Diagnose what your model is missing** by reading residuals
5. **Compare different regression models** (Linear vs Random Forest)
6. **Improve a model** by adding features and per-class approaches

This is the same workflow as classification. The only difference: **you're predicting a number, not a category.**

Which Approach Should I Use?

	Classification	Regression
Question	"What is this?"	"How much?"
Output	A category (STAR, GALAXY, QSO)	A number (z-magnitude, temperature)
Example	Classify a spectrum as a star or quasar	Predict the brightness of that star

Your research question determines your approach.

We are doing supervised learning. **Unsupervised learning**, **clustering**, and **dimensionality reduction** are different tools for different questions.

When Do We Use Regression?

Beyond classification, we often need to make **continuous predictions**:

Problem	Features	Target
Stellar brightness	Colors (u, g, r, i, z)	Magnitude (0–30)
Molecular energy	Atomic structure	Bond energy (continuous)
Star temperature	Spectral absorption lines	Temperature (K)
Galaxy redshift	Brightness + colors	Redshift z (0–7)

Key insight: When you need a precise number, not just a category, regression is your tool.

Linear Regression: The Simplest Model

Fit a straight line (or plane, or hyperplane) through the data.

Prediction formula:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

Where:

- $\hat{y}$ = predicted value
- x_i = features (colors, spectrum)
- β_i = learned weights (how much each feature matters) - often collected as a "weights vector": β

Goal: Find coefficients β_i that minimize error. You are free to choose how to measure error, but the most common choice is **Mean Squared Error (MSE)**.

Measuring Error in Regression

What is a residual?

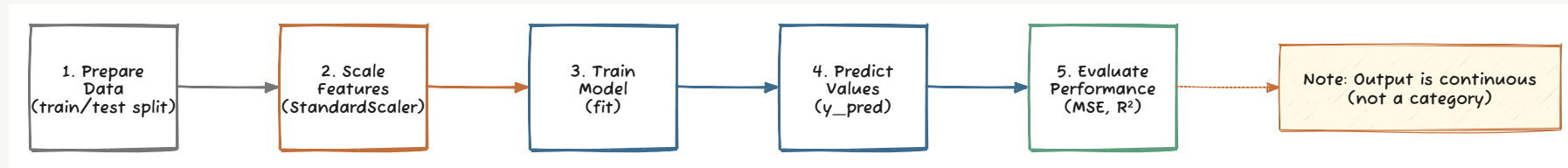
$$\text{residual} = \text{predicted} - \text{actual}$$

Large residual = bad prediction. Small residual = good prediction.

Why minimize squared error (MSE)?

- Penalizes big mistakes harder than small ones
- Mathematically convenient (smooth, differentiable)
- Interpretable: MSE is in **squared** units of your target (e.g., if predicting magnitude, MSE is in magnitude^2).

The Regression Pipeline



Identical to classification: Prepare → Scale → Train → Predict → Evaluate

The only difference: your target is continuous, not categorical.

Building the Model (Still the Same Pattern)

The `scikit-learn` workflow is **identical to KNN**:

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# 1. Prepare: split the data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# 2. Scale: fit on training data only
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```


Building the Model (Still the Same Pattern)

The `scikit-learn` workflow is **identical to KNN**:

```
# 3. Train & Predict
model = LinearRegression()
model.fit(X_train_scaled, y_train)
y_pred = model.predict(X_test_scaled)

# 4. Evaluate
from sklearn.metrics import mean_squared_error, r2_score
print(r2_score(y_test, y_pred))
```

Same structure, every time.

Evaluation: MSE and R²

Mean Squared Error (MSE) — average squared distance between predicted and actual

$$\text{MSE} = \frac{1}{n} \sum (y_{\text{true}} - y_{\text{pred}})^2$$

Lower is better. In same units as your squared target.

R² (R-squared) — fraction of variance explained

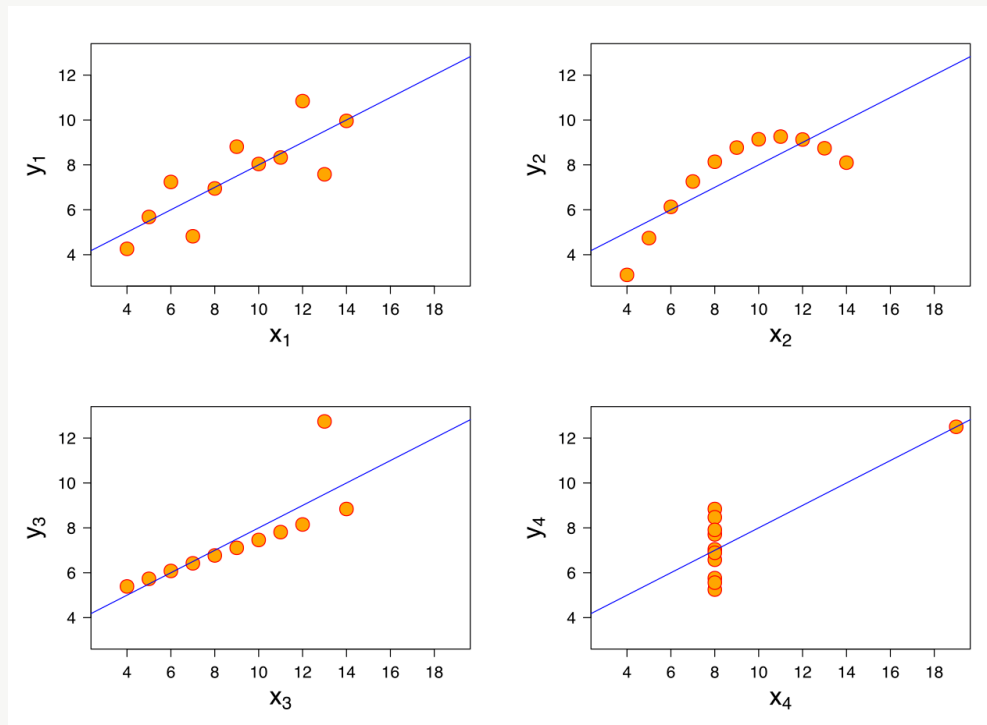
$$R^2 = 1 - \frac{\text{SS}_{\text{res}}}{\text{SS}_{\text{tot}}}$$

where $\text{SS}_{\text{tot}} = \sum (y_{\text{true}} - \bar{y})^2$ is the total variance in the data.

- 1.0 = perfect fit; 0.95 = explains 95% of variance; 0.5 = explains 50%
- 0.0 = no better than mean prediction

A Warning: Anscombe's Quartet

Four datasets. Identical means, variances, correlations, AND R^2 values.

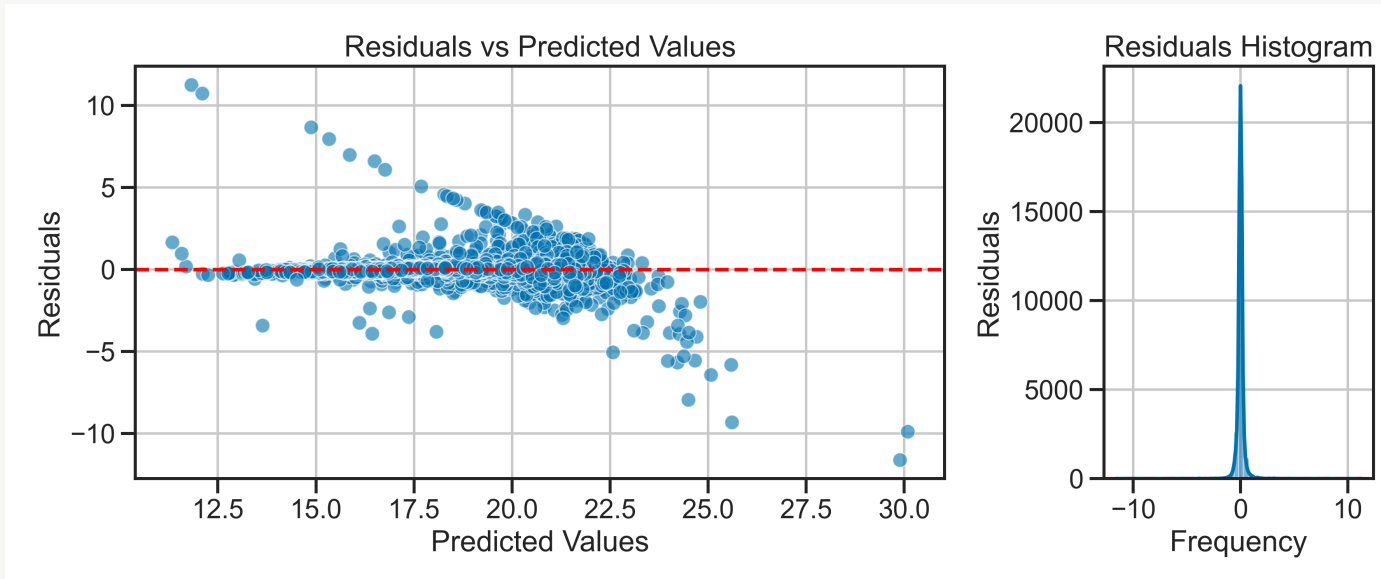


Identical statistics **do not** mean identical data. Always plot. Source: https://en.wikipedia.org/wiki/Anscombe's_quartet

Diagnostic Plots: Reading Residuals

Always visualize three things:

1. **Residuals vs Predicted** — scatter plot: do residuals look random?
2. **Histogram of residuals** — distribution: is it centered at zero?
3. **Actual vs Predicted** — 45° line should align with your points



Residuals Diagnostic Guide	
Pattern	What It Means
✓ Random scatter around zero	Model is good No systematic error
✗ Curved pattern	Non-linear relationship Linear model is wrong
✗ Fan shape (variance increases)	Heteroscedasticity Model less accurate at high values

Rule: Residuals should look like **random noise** around zero.

If they have a pattern (curved, fanned, clustered), your model is missing something.

When Residuals Tell You Something

Your model doesn't fit

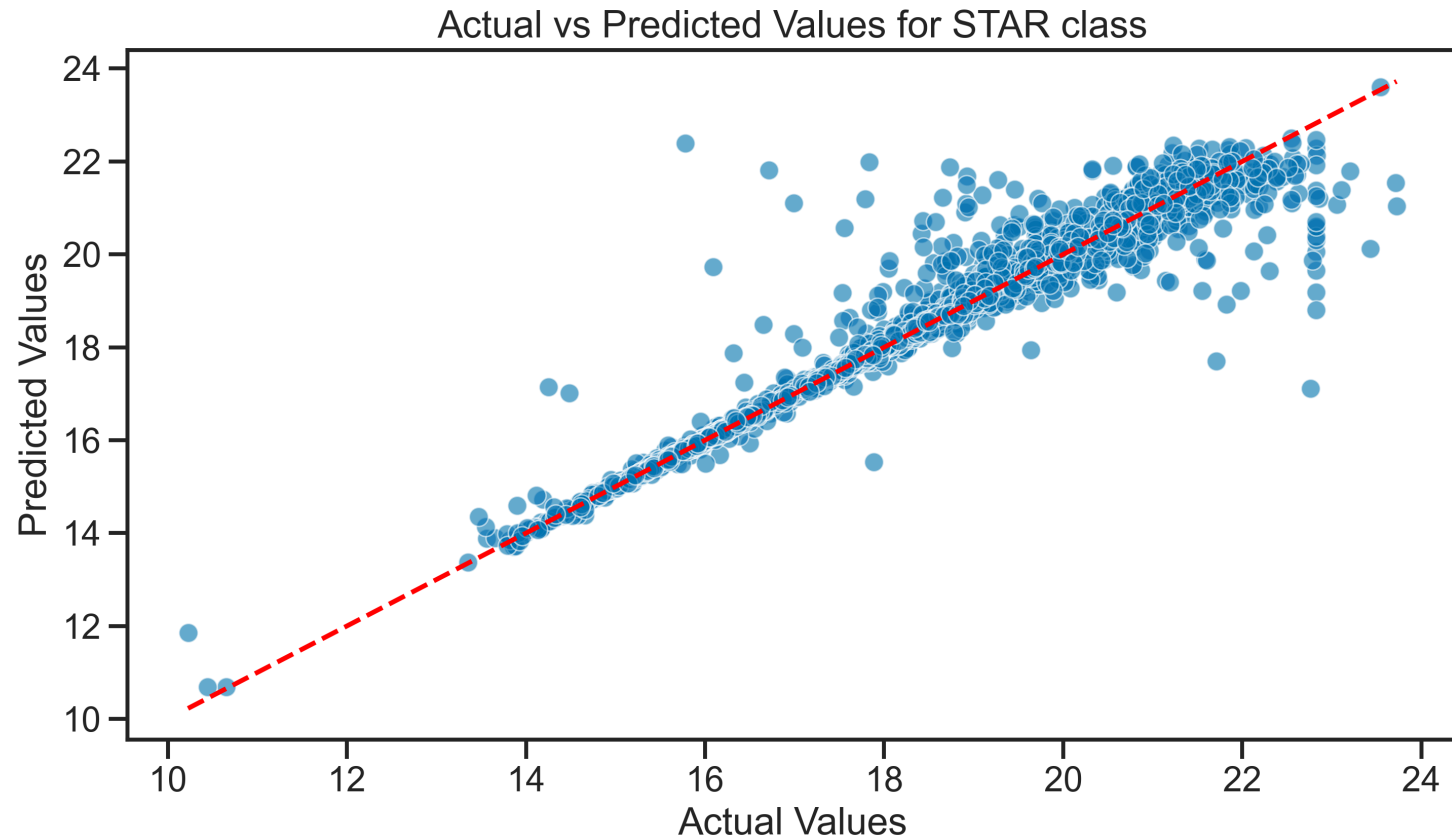
- **Curved pattern** → Relationship is non-linear (not a straight line)
- **Fan shape** → Error gets bigger at high values (heteroscedasticity)
- **Clustering** → Model works better for some groups than others

What to do?

- Try a **different model** (Random Forest, polynomial)
- **Add more features** (maybe colors interact?)
- **Separate by class** (STAR and GALAXY have different color-brightness relationships)

When Linear Isn't Enough

Same data, same question. Different model.



When Linear Isn't Enough

Random Forest can capture **non-linear patterns** — curves, interactions, complexity that linear models miss.

Same `scikit-learn` API:

```
from sklearn.ensemble import RandomForestRegressor

model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train_scaled, y_train)
y_pred = model.predict(X_test_scaled)
```


Linear vs Random Forest Regression

Criterion	Linear Regression	Random Forest
Speed	Fast ✓	Slower ✗
Interpretability	High (see β coefficients)	Low (black box)
Flexibility	Low (straight line)	High (curves & interactions)
Overfitting Risk	Low	High (needs tuning)

Model Comparison: Linear vs Random Forest

Linear regression assumes a straight line. Fast, interpretable, fails on curves.

Random Forest is flexible but slower and prone to overfitting.

Which to use? Visualize your residuals first. If they're curved, try Random Forest. If they're random, Linear might be enough.

Does Redshift Dominate Here Too?

Without redshift	With redshift
Low R^2 (colors alone are weak predictors)	High R^2 (redshift correlates with magnitude)
Colors matter	But is it "cheating"?

Key question: Adding redshift improves R^2 , but does the model learn color-magnitude relationships, or just memorize redshift?

Your job: Compare residuals with and without redshift. Does one have obvious patterns?

Today's Activity: Regression with scikit-learn

Work through **Activity 03** step by step:

Part 1: Baseline Linear Model

- Load SDSS data (100k objects, 17 features)
- Predict **z-band magnitude** from colors (u, g, r, i)
- Evaluate with **MSE, R^2**

Part 2: Diagnose with Residuals

- Plot residuals vs predicted
- Is there a pattern? Curved? Fanned out?
- What does that tell you?

Today's Activity: Regression with scikit-learn

Work through **Activity 03** step by step:

Part 3: Add Information

- Add **redshift** as a feature
- Compare R^2 before and after
- Do residuals look better? Why?

Part 4: Different Approach

- Build **per-class models** (STAR, GALAXY, QSO separately)
- Compare accuracy per class
- Trade-off: flexibility vs sample size per model

Today's Activity: Regression with scikit-learn

Work through **Activity 03** step by step:

Part 5: Beyond Linear

- Try **Random Forest Regression**
- Compare performance to Linear
- Why might it be better? Overfitting risk?

The goal: Understand the whole workflow. Pick one question that interests you and dig deep.

Notes: [The Importance of Visualization](#) · [Methods & Validation](#)